



DANI CABALLERO MÉLICH

AI / AGENTIC SYSTEMS ENGINEER

Barcelona · remote-first
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STACK

Agents LangGraph · MCP ·
Pydantic · multi-agent
orchestration · human-in-the-
loop gating · guardrails

Retrieval RAG · hybrid dense +
lexical · embeddings · sqlite-vec ·
recall@k · MRR

Evaluation validator suites ·
scored thresholds · output repair

Providers Anthropic · OpenAI ·
per-stage model and effort
routing · fallback chains · cost
and token telemetry

LANGUAGES

Python 3.12 · C# · Kotlin ·
TypeScript · GDScript

SYSTEMS

SQLite · JSON-RPC · asyncio ·
pytest · Rust/Tauri · React · Git

ALSO

Three years shipping production
software fully remote, cross-
timezone with a US studio.

Harness source available on
request; unpublished while I
explore commercialising it.

I build LLM pipelines meant to survive production: typed contracts, human gates, retrieval scored against a gold set, and a cost line on every stage.

EXPERIENCE

Independent · Software & AI Systems

Jun 2025 – present

- A 19-node LangGraph pipeline: 10 role-specialised agents, 5 human approval gates.
- It turns a written brief into validated code that ships in a commercial mobile game.
- Typed Pydantic contracts are the only source of truth; every other view is generated.
- Evaluation harness: 9 validators plus scored visual comparison, gated at 0.85 / 0.70.
- Failures route back by conditional edge to the stage that caused them.
- Hybrid retrieval on a local sqlite-vec index, scored against a 32-pair gold set.
- Recall@10 of 0.969, MRR 0.66. Cost telemetry per stage: **\$0.95 an execution.**
- ~15,300 lines of Python 3.12, 259 tests. Durable on SQLite. Exposed as an MCP server.

Somni Game Studios · Software Engineer

Jul 2025 – Mar 2026 · Stepland, live on Google Play and the App Store

- Drove the coding-agent sessions that rebuilt the tutorial: 82 declarative steps.
- About 76% of the shipped system, against a fixed trade-show date.
- Replaced a silently-failing third-party dependency with an in-house Kotlin plugin.
- Wrote it in a week, in a language I had not used. 211 of its 212 lines.
- Owned app boot, cloud sync and remote config. 7 of the 16 services are mine.

Freelance · Contract development and art

Oct 2024 – Jun 2025

Purple Pwny Studios · Software Engineer

Aug 2023 – Oct 2024 · remote, US studio · three people

- Built an expert system: a heuristic policy run under Monte Carlo simulation, used as one reference point for automated economy balancing. Rule-based, not machine learning.
- Built a real-time client end to end, and the bridge marshalling live state to the web.

PROJECT

Dungeon Heroes · solo · closed testing on Google Play

- Real-time action roguelike. ~20,100 lines of C#, 102 abilities, 24 UI screens.
- Versioned save-schema migrations, cloud sync, live-ops SDKs, signed release builds.

Architecture write-up, telemetry and code walkthrough at danimelich.com